Aman Kumar

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Academic Appointments

Sept. 2026– present **Postdoctoral Fellow**
Nanyang Technological University, Singapore
Advisor: Prof. Justin Song

Sept. 2023– Sept. 2026 **Dirac Postdoctoral Fellow, Physics**
National High Magnetic Field Laboratory, Tallahassee, Florida
Advisor: Prof. Hitesh Changlani

Education

2018–2023 **Ph.D. in Physics**
Tata Institute of Fundamental Research, Mumbai, India
Advisor: Prof. Vikram Tripathi

2016–2018 **M.Sc. in Physics**
Tata Institute of Fundamental Research, Mumbai, India

2013–2016 **B.Sc. (Hons.) in Physics**
University of Delhi, India

Academic Research Visit

Sept 1st - Nov 1st 2022 **Visit to Prof. Subir Sachdev's group**
Harvard University, Cambridge, USA

Awards and Fellowships

Sept 2023 **Dirac Fellowship**
National High Magnetic Field Laboratory

Mar 2022 **Sarojini Damodaran International Travel Fellowship**
Tata Institute of Fundamental Research, Mumbai, India

Mar 2022 **APS Distinguished Student Award**

Mar 2016 **Academic Merit Award in B.Sc.**
A.R.S.D. College, University of Delhi

Areas of Research

- Wigner crystal physics in triangular moiré systems
- Study of emergent phenomena such as fractionalization and topological order
- Quasiparticle stability in quantum many-body systems such as frustrated magnets (Kitaev systems) and underdoped cuprates
- Formulation of thermal Hall transport using tensor-network methods
- Many-body localization in Fock space

Skills

- Programming: Python, C, C++, Bash

- High-performance computing
- Tensor-network techniques
- Large-scale exact diagonalization, FEAST method
- Familiarity with many-body field theory techniques

Teaching Assistant Experience

	Courses	**Instructor**
Jan-July. 2020	Statistical Physics I	Kedar Damle
Sept-Jan 2020-2021	Condensed Matter Field Theory	Kedar Damle
Jan-July. 2021	Statistical Physics II	Tridib Sadhu
Jan-July. 2023	Quantum Mechanics II	Vikram Tripathi

Publications

- **Melting temperature shifts from quantum fluctuations in generalized Wigner crystals**, *Aman Kumar, Sogoud Sherif, Veit Elser and Hitesh J. Changlani*, **arXiv:2604.19870**
- **Origin and stability of generalized Wigner crystallinity in triangular moiré systems**, *Aman Kumar, Cyprian Lewandowski, Hitesh Changlani*, **npj Quantum Materials 10, 95** (2025)
- **Haerter-Shastry kinetic magnetism and metallicity in the triangular Hubbard model**, *Sogoud Sherif, Prakash Sharma, Aman Kumar, Hitesh J. Changlani*, **Commun Phys 9, 318 (2026)**
- **Identifying chiral topological order in microscopic spin models by modular commutator**, *Avijit Maity, Aman Kumar, Vikram Tripathi*, **arXiv:2510.06086**
- **Field tuning Kitaev systems for spin fractionalization and topological order**, *Jagannath Das, Sarbajaya Kundu, Aman Kumar, Vikram Tripathi*, **J. Physics Condens. Matter** 36 443001(2024)
- **Thermal Hall conductivity near field-suppressed magnetic order in a Kitaev-Heisenberg model**, *Aman Kumar, Vikram Tripathi*, **Phys. Rev. B** 107, L220406 (2023)
- **Quasiparticle metamorphosis in the random t-J model**, *Aman Kumar, Subir Sachdev, Vikram Tripathi*, **Phys. Rev. B** 106, L081120 (2022)
- **Kitaev quasiparticles in a proximate spin liquid: A many-body localization perspective**, *Aman Kumar, Vikram Tripathi*, **Phys. Rev. B** 102, 100401(R) (2020)
- **Jordan-Wigner fermionization of quantum spin systems on arbitrary two-dimensional lattices: A mutual Chern-Simons approach**, *Jagannath Das, Aman Kumar, Avijit Maity, and Vikram Tripathi*, **Phys. Rev. B** 108, 085110 (2023)
- **Microscopic theory of field-tuned topological transitions in the Kitaev honeycomb model**, *Jagannath Das, Aman Kumar, and Vikram Tripathi*, **Phys. Rev. B** 111, L041111 (2025)

Conferences and Talks

Mar 19, 2026 **Quantum and classical thermal melting of generalized Wigner crystals**, *APS March Meeting*

Aug 29, 2025 **Finite-temperature study of generalized Wigner crystals**, *Random Interaction Seminar, TIFR Mumbai*

Mar 19, 2025 **Origin and stability of generalized Wigner crystallinity in triangular moiré systems**, *APS March Meeting*

Aug 21, 2024 **Origin and stability of generalized Wigner crystallinity in triangular moiré systems**, *Seminar at IIT Bombay*

Aug 8, 2024 **Origin and stability of generalized Wigner crystallinity in triangular moiré systems**, *Random Interaction Seminar, TIFR Mumbai*

Oct 7, 2022 **Thermal Hall conductivity near field-suppressed magnetic order in a Kitaev-Heisenberg model**, *Harvard University*

Dec 11, 2021 **Quasiparticle stability in the hole-doped random t-j model**, *QMAT National Conference on Quantum Condensed Matter*

Posters Presented

Aug 2–14, 2025 **Origin and stability of generalized Wigner crystallinity in triangular moiré systems**, *Workshop on Advances in Quantum Matter*
ICTP, Trieste, Italy

Jan, 6-10, 2025 **Origin and stability of generalized Wigner crystallinity in triangular moiré systems**, *2025 Theory Winter School, Maglab*
NHFML, Tallahassee, Florida

Dec 3–10, 2023 **Thermal Hall conductivity near field-suppressed magnetic order in a Kitaev-Heisenberg model**, *Conference on Fractionalization and Emergent Gauge Fields in Quantum Matter*
ICTP, Trieste, Italy

Jun 20–25, 2022 **Quasiparticle metamorphosis in the hole-doped random t-j model**, *International Conference on Highly Frustrated Magnetism (HFM)*
Paris, France

Apr 15–20, 2021 **Kitaev quasiparticles in a proximate spin liquid: A many-body localization perspective**, *Korrelationstage 2021 (online)*

Feb 17–28, 2020 **Kitaev quasiparticles in a proximate spin liquid: A many-body localization perspective**, *Gapless Fermions – from Fermi liquids to strange metals*
Max Planck Institute for the Physics of Complex Systems (MPIPKS), Dresden (online)

Dec 23, 2019 – Jan 2, 2020 **Kitaev quasiparticles in a proximate spin liquid: A many-body localization perspective**, *Novel Phases of Quantum Matter*
ICTS Bangalore, India

Nov 11–29, 2019 **Kitaev quasiparticles in a proximate spin liquid: A many-body localization perspective**, *Thermalization, Many-body Localization, and Hydrodynamics*
ICTS Bangalore, India

References

Prof. Vikram Tripathi
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Prof. Hitesh Changlani
Department of Physics,
Florida State University, Tallahassee, Florida
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Prof. Subir Sachdev
Department of Physics
Harvard University, MA, US, 02138
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Prof. Veit Elser
Department of Physics,
Cornell University, Ithaca, NY
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